Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P2-Model)** **Date: 03-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-29** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**4**	2	**1**	3	**4**	4	**6**	5	**5**	6	**2**
7	**1**	8	**7**	9	**C**	10	**ABCD**	11	**ACD**	12	**ABCD**
13	**ABD**	14	**ABC**	15	**B**	16	**B**	17	**D**	18	**A**

PHYSICS

19	**4**	20	**3**	21	**6**	22	**3**	23	**4**	24	**4**
25	**6**	26	**4**	27	**BD**	28	**ABC**	29	**BD**	30	**BC**
31	**ACD**	32	**BCD**	33	**A**	34	**B**	35	**C**	36	**D**

CHEMISTRY

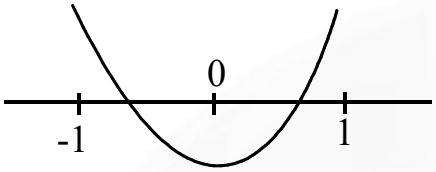
37	**6**	38	**9**	39	**4**	40	**8**	41	**2**	42	**9**
43	**2**	44	**5**	45	**BCD**	46	**ABC**	47	**ABCD**	48	**BD**
49	**BC**	50	**CD**	51	**C**	52	**A**	53	**B**	54	**D**

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SOLUTIONS
MATHEMATICS

1. $3x^2+x-1=4\left|x^2-1\right|$

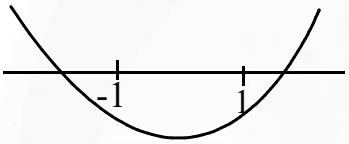
If $x\in[-1,1]$, $3x^2+x-1=-4x^2+4 \Rightarrow 7x^2+x-5=0$



Say $f(x)=7x^2+x-5$

$f(1)=3; f(-1)=1; f(0)=-1$ [Two Roots]

If $x\in(-\infty,-1]\cup[1,\infty)$,



$3x^2+x-1=4x^2-4 \Rightarrow x^2-x-3=0$

Say $g(x)=x^2-x-3$

$g(-1)=-1; g(1)=-3$ [Two roots]

2. $\sqrt{3}\cos x+\frac{2b}{a}\sin x=\frac{c}{a}$

Now, $\sqrt{3}\cos\alpha+\frac{2b}{a}\sin\alpha=\frac{c}{a}$ .....(1)

$\sqrt{3}\cos\beta+\frac{2b}{a}\sin\beta=\frac{c}{a}$ ....(2)

$\sqrt{3}\left[\cos\alpha-\cos\beta\right]+\frac{2b}{a}\left(\sin\alpha-\sin\beta\right)=0$

$\sqrt{3}\left[-2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)\right]+\frac{2b}{a}\left[2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)\right]=0$

$-\sqrt{3}+2\sqrt{3}.\frac{b}{a}=0$

$\frac{b}{a}=\frac{1}{2}=0.5$

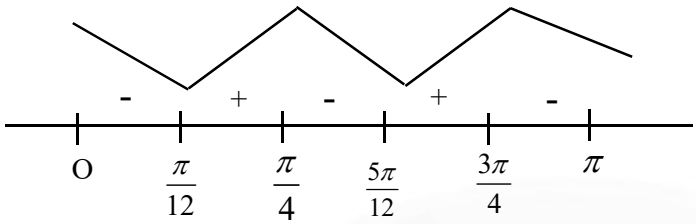
3. $f(\theta)=(\sin\theta+\cos\theta)^2+(\sin\theta-\cos\theta)^4$

$f(\theta)=\sin^2 2\theta-\sin 2\theta+2$

$f'(\theta)=2(\sin 2\theta).(2\cos 2\theta)-2\cos 2\theta=2\cos 2\theta(2\sin 2\theta-1)$

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critical points



So, minimum at $\theta = \frac{\pi}{12}, \frac{5\pi}{12}$

$\lambda_1 + \lambda_2 = \frac{1}{12} + \frac{5}{12} = \frac{6}{12} = \frac{1}{2}$

4. $5 = \frac{3+7+x+y}{4} \Rightarrow x+y=10$

$\text{var}(x) = 10 = \frac{3^2+7^2+x^2+y^2}{4} - 25$

$140 = 49 + 9 + x^2 + y^2$

$x^2 + y^2 = 82$

$x + y = 10$

$\Rightarrow (x, y) = (9,1)$

Four numbers are 21, 9, 10, 8

Mean $= \frac{48}{4} = 12$

5. $\underbrace{(1+3+5+....+a)}_{let\ n_1\ terms} + \underbrace{(1+3+5+....+b)}_{n_2\ terms} = \underbrace{(1+3+5+....+c)}_{n_3\ terms}$

$\Rightarrow {n_1}^2 + {n_2}^2 = {n_3}^2$, where $a = 2n_1 - 1,\ b = 2n_2 - 1,\ c = 2n_3 - 1$

$a + b + c = 57 \Rightarrow (2n_1 - 1) + (2n_2 - 1) + (2n_3 - 1) = 57 \Rightarrow (n_1 + n_2 + n_3) = 30$

$\Rightarrow n_1 = 5, n_2 = 12, n_3 = 13$

$\Rightarrow a = 9,\ b = 23,\ c = 25$

6. $\frac{1}{n\pi}\int_{-n}^{n} \frac{dx}{\sqrt{\{x\}\{-x\}}} = \frac{1}{n\pi}.2n\int_0^1 \frac{dx}{\sqrt{\{x\}\{-x\}}} = \frac{2}{\pi}\int_0^1 \frac{dx}{\sqrt{x(1-x)}} = 2$

7. $\cos^{-1}x \in \left(0, \frac{\pi}{2}\right) \Rightarrow \sin^{-1}x = \frac{\pi}{2} - \cos^{-1}x \in \left(0, \frac{\pi}{2}\right)$

$\therefore \sin\left(\cos^{-1}x\right) = \cos\left(\sin^{-1}x\right) = \sqrt{1-x^2}$

8. We have $\frac{\cos 8x - \cos 7x}{1 + 2\cos 5x} = \cos 3x - \cos 2x$

$\Rightarrow \int \frac{\cos 8x - \cos 7x}{1 + 2\cos 5x} dx = \int (\cos 3x - \cos 2x) dx = \frac{1}{3}\sin 3x - \frac{1}{2}\sin 2x + C$

Here, we got $p = \frac{1}{3}, q = \frac{-1}{2}$

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9. $A_{m,n}^{-1}=\dfrac{adj(A_{m,n})}{\det(A_{m,n})}$ where $adj(A_{m,n})=\begin{pmatrix} mn-1 & 1-m & 1-n \\ 1-m & m^2-1 & 1-m \\ 1-n & 1-m & mn-1 \end{pmatrix}$

$\because m=1\Rightarrow |A|=0,$ we take $m\neq 1$

$|A|=m(mn-1)-1(m-1)+1(1-n)=m^2n-2m-n+2=(m-1)(mn+n-2)$

$$m=0\Rightarrow adj(A_{m,n})=\begin{pmatrix} -1 & 1 & 1-n \\ 1 & -1 & 1 \\ 1-n & 1 & -1 \end{pmatrix},|A|=(-1)(n-2)$$

The value of n can be 1

$$m=-1\Rightarrow adj(A_{m,n})=\begin{pmatrix} -n-1 & 2 & 1-n \\ 2 & 0 & 2 \\ 1-n & 2 & -n-1 \end{pmatrix},|A|=(-2)(-2)=4$$

10. $\hat{a},\hat{b},\hat{c}$ form an equilateral triangle and angle between each pair is $\dfrac{2\pi}{3}$

$$\therefore \left|x\bar{a}+y\bar{b}+z\bar{c}\right|^2=x^2+y^2+z^2-xy-yz-zx=\frac{1}{2}\left[(x-y)^2+(y-z)^2+(z-x)^2\right]$$

Minimum value $=\dfrac{1}{2}[1+1+4]=3$

$\therefore \left|x\bar{a}+y\bar{b}+z\bar{c}\right|\geq\sqrt{3}$

11. If $f(x)$ is continuous at $x=0$.

$f(0)=\lim\limits_{x\to 0} f(x)$

$$0=\lim_{x\to\infty} x^m\left(\cos x\cos\frac{1}{x}+\sin x\sin\frac{1}{x}\right)$$

$$0=\lim_{x\to\infty} x^m\left(x^m\cos x\cos\frac{1}{x}+\sin\left(\frac{\sin x}{x}\right)\sin\frac{1}{x}\right)$$

true when $m>0$

If $f(x)$ is differentiable at $x=0$

$\lim\limits_{h\to 0}\dfrac{f(h)-f(0)}{h}$ exist

$$\lim_{h\to 0}\frac{h^m\left(\cosh\cos\frac{1}{h}+\sinh\sin\frac{1}{h}\right)-0}{h}$$ exist

12. Let $y=mx+c$ is common tangent

$$\therefore c=-m^2=\pm\sqrt{\frac{m^2}{4}+\frac{3}{4}}=\frac{a}{m}$$

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$\Rightarrow m=-1, a=1, c=-1$

Common tangent $y=-x-1$

$x+y+1=0$

$l=1,\ m=1$

13. at least one fruit is chosen.

$(4+1)(3+1)(2+1)(2+1)-1=179$

at least 2 fruits are chosen.

$(4+1)(3+1)(2+1)(2+1)-1-(4)=175$

At most 2 fruits can be chosen

Selection of No fruit $\Rightarrow 1$ method

Selection of 1 fruit $\Rightarrow 4$ method

Selection of 2 fruits $\Rightarrow 4+{}^{4}C_{2}=10$

$=1+4+10=15$

14. Equation of P_1

$2(x-1)-2(y+1)+2(z-1)=0$

$\Rightarrow x-y+z-3=0 \quad \ldots\ldots(1)$

Equation of P_2 is

$(-2)(x-1)+3(y-2)+2(z-1)=0$

$\Rightarrow 2x-3y-2z+6=0 \quad \ldots\ldots(2)$

Vector along line of intersection of plane P_1 *and* P_2 is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & -3 & -2 \end{vmatrix} = 5\hat{i}+4\hat{j}-\hat{k}$$

dr of line is $5,4,-1$

in (1) and (2) put $z=0$, we get

$x=15, y=12$

Equation of line of intersection of P_1 *and* P_2 is

$$\frac{x-15}{5}=\frac{y-12}{4}=\frac{z-0}{-1}$$

$P_3: P_1+\lambda P_2=0$

$(x-y+z-3)+\lambda(2x-3y-2z+6)=0$

Passes through $(1,0,-1)$

$\Rightarrow 16x-19y+4z=12$

$\therefore$ length of $\perp$ from $(0,0,0)$ to P_3 is $=\dfrac{12}{\sqrt{633}}$

15. Multiply R_2 by z and R_3 by y and use $R_1 \to R_1-R_2+R_3$

16. $A=\{HH, HTHH, HTHTHH, \ldots\}$

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And $B=\{TT,THTT,THTHTT,....\}$ $P(A)=\frac{1}{2^2}+\frac{1}{2^4}+.....=\frac{1}{3}$; $P(B)=\frac{1}{3}$

Required Probability $=\frac{1}{3}+\frac{1}{3}=\frac{2}{3}$

17. $\sum_{k=1}^{100} i^{k!}+\sum_{k=1}^{100}\omega^{k!}$

$\sum_{k=1}^{100} i^{k!}=i^{1!}+i^{2!}+i^{3!}+i^{4!}+.....+i^{100!}$

$=i-1+i^6+1+1+1+....+1=i-2+97$

$=i+95$

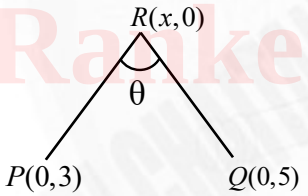
$\sum_{k=1}^{100}\omega^{k!}=\omega^1+\omega^{2!}+\omega^{3!}+\omega^{4!}+.....+\omega^{100!}$

$=\omega+\omega^2+1+1+1+.....+1=97$

sum $=i+95+97=i+192$

18. $m_1=-\frac{3}{x}$

$m_2=-\frac{5}{x}$



$\tan\theta=\left|\frac{m_1+m_2}{1+m_1m_2}\right|=\left|\frac{\frac{2}{x}}{1+\frac{15}{x^2}}\right|$

$\tan\theta=2\left|\frac{x}{15+x^2}\right|$

$\lambda=\frac{x}{15+x^2}$

$\frac{d\lambda}{dx}=\frac{(15+x^2)-2x^2}{(15+x^2)^2}=0$

$x^2=15\Rightarrow|x|=\sqrt{15}$

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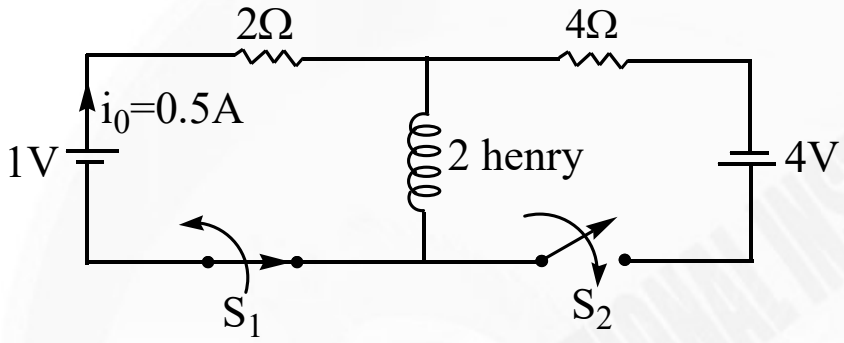
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19. Balance this force a difference in the mercury level's is setup. Let h be the difference in level $BIa = (\rho gh) \times a^2$

$$h = \frac{BI}{\rho ga} = \frac{5.44 \times 10}{13.6 \times 10^3 \times 10 \times 4 \times 10^{-3}}$$

$$= \frac{1}{10} = 0.1m = 100\,mm$$

20.



$i_0 = 0.5A$

When; S_2 is closed;

$$-L\frac{di}{dt} - 4i + 4 = 0$$

$$-2\frac{di}{dt} - 4i_0 + 4 = 0$$

$$-2\frac{di}{dt} + 4 \times 0.5 + 4 = 0 \qquad [\because i_0 = -0.5A]$$

$$\frac{di}{dt} = 3$$

21. At half power frequencies,

$$Z = R\sqrt{2} \Rightarrow |X_L - X_C| = R$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2 \Rightarrow \omega^2 L^2 + \frac{1}{\omega^2 C^2} - \frac{2L}{C} = R^2$$

$$\omega^2 L^2 C^2 - \omega^2 \left(\frac{2L}{C} + R^2\right) C^2 + 1 = 0$$

The product of roots is

$$\omega_1^2 \omega_2^2 = \frac{1}{L^2 C^2} \Rightarrow \omega_1 \omega_2 = \frac{1}{LC} = \omega_0^2$$

$$\Rightarrow \omega_0 = \sqrt{\omega_1 \omega_2} = 60\,rad/s \Rightarrow \omega_0 / 10 = 6$$

22. Wavelength gap: $\left|\lambda_{cut\ off} - \lambda_{k\alpha}\right|$

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$$= \lambda_{K_a} - \frac{hc}{ev_a}$$

$$\left[\lambda_{K_a} - \frac{hc}{e(20)}\right] = \eta\left[\lambda_{K_a} - \frac{hc}{e(10)}\right]$$

$$\lambda_{K_\alpha}[1-\eta] = \eta\left[\frac{hc}{e(10)} - \frac{hc}{e(10)}\eta\right] \times 10^{-3}$$

$$= \frac{hc}{e}\left[\frac{1}{20} - \frac{\eta}{10}\right] \times 10^{-3}$$

$$\frac{hc}{\lambda_{K_\alpha}} = 13.6\left(\frac{1}{1^2} - \frac{1}{2^2}\right)(28)^2$$

$$\frac{hc}{\lambda_{K_\alpha}} = 13.6 \times \frac{3}{4} \times 28^2$$

$$\lambda_{K_\alpha} = \frac{hc \times 4}{13.6 \times 3 \times 28^2}$$

$$\frac{4hc}{13.6 \times 3 \times 28^2 e}[1-\eta] = \frac{hc}{e}\left[\frac{1}{20} - \frac{h}{10}\right] \times 10^{-3}$$

$$\frac{1}{8}[1-\eta] = \left(\frac{1}{20} - \frac{\eta}{10}\right)$$

$$\frac{1-\eta}{2-\eta} = \frac{8}{10} = \frac{4}{5}$$

$$\eta = 3$$

23. The given ray cuts y-axis at (0, 1) and x-axis at (3, 0). Taking u = +3 cm, f = +15 cm, we get v = +2.5 m. So, the refracted ray passes through (0, 1) and (2.5, 0). Its equation then becomes y = –0.4x +1.

24. $\sin\theta = \frac{1}{d}\left[\Delta x - (l_2 - l_1)\right] = \frac{1}{d}\left(n\lambda - (l_2 - l_1)\right]$

$$= \frac{1}{10 \times 10^{-6}}\left[n \times 500 \times 10^{-9} - 20\right]$$

$\sin\theta = 2\left[\frac{n}{40} - 1\right]$ Hence $\theta = \sin^{-1}\left[2\left(\frac{n}{40} - 1\right)\right]$

$|\sin\theta| \le 1$

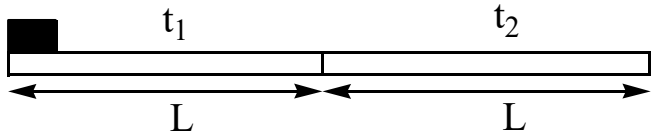
$$-1 \le 2\left[\frac{n}{40} - 1\right] \le 1$$

$$-20 \le [n - 40] \le 20$$

$$20 \le n \le 60$$

Hence number of maxima = 60 – 20 = 40.

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25.

$a_b = -\mu g$

$2L = \frac{1}{2}(\mu g)(t_1 + t_2)^2$

$L = O \times t_2 + \frac{1}{2}\mu g t_2^2$

$t_2 = \sqrt{\frac{2L}{\mu g}}$

$t_1 = (2 - \sqrt{2})\sqrt{\frac{L}{\mu g}}$

$\frac{\mu mg}{2M} = a_1; a_2 = \frac{\mu mg}{M}$

$v_1 = \frac{\mu mg}{2M} \times t_1$

$v_2 = v_1 + \frac{\mu mg}{M} t_2$

$= \frac{mg}{2M} t_1 + \frac{\mu mg}{M} + t_2$

$\frac{v_2}{v_1} = 1 + \frac{2t_2}{t_1} = \frac{\sqrt{2}+1}{\sqrt{2}-1}$

26. $\lambda = 1.57$

$\omega_b = \Delta V_E + \Delta U_{surface}$

$\frac{\varepsilon_r \varepsilon_0 A V^2}{d} = \frac{1}{2}\frac{\varepsilon_r \varepsilon_0 A V^2}{d} + \sigma \cdot 2A$

$V = \sqrt{\frac{4\sigma d}{\varepsilon_0 \varepsilon_r}}$

27. $W_{man} = \Delta KE$

From momentum conservation $mv_{man} = Mv$

For [a]: Work done by man on boat $= \Delta KE_{boat} = \frac{1}{2}Mv^2$

For [b]: $W_{man\ total} = \Delta KE_{system} \frac{1}{2} m \frac{M^2 v^2}{m^2} + \frac{1}{2} m v^2$

For [c]: $v_{cm} = 0$ {momentum conservation}

For [d]: $W_{man\ on\ himself} = \Delta KE_{man} = \frac{1}{2}\frac{M^2}{m} v^2$

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28. $U = 2x^3 - 9x^2 + 12x$

$F = -\frac{dU}{dx} = \left(-6x^2 + 18x - 12\right)$

For equilibrium F = 0, x = 1 and 2

At x = 0, F = –12N

$\frac{d^2U}{dx^2} = 12x - 18$

At x = 1, $\frac{d^2U}{dx^2} = -6$

Maxima x = 2, $\frac{d^2U}{dx^2} = +6$

Minima

$U_{max} = 2(1)^3 - 9\times1^2 + 12\times1 = 5J$

$U_{min} = 2\cdot(2)^3 - 9\times(2)^2 + 12\times2 = 4J$

For oscillatory motion 4J < M.E. < 5J

29. As in supercooled state specific heat of system is $= 4.2KJ\,kg^{-1}K^{-1}$

$1\times4.2\times10 = m334$

$m = \frac{4.2}{334} \cong 0.13kg$

Latent heat will remain same

30. $\Delta Q = \frac{5}{2}nR\times\left[\frac{P_0V_0 - P_0V_i}{nR}\right]$

$V_i = 8\times10^{-3}$

$V_g = \left(8+2\times2\right)\times10^{-3}$

$\Delta Q = 1000J$

$P_B = P_0 + \frac{kx}{A}$

$= 10^5 + \frac{2000\times2\times10^{-1}}{2\times10^{-2}} = 1.2\times10^5$

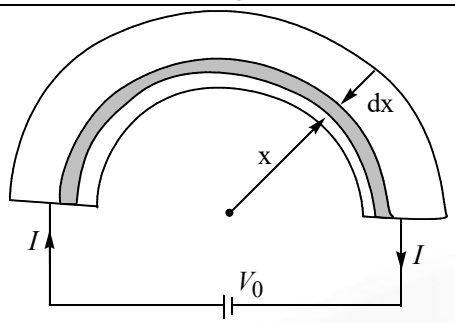
Up to A, $W_1 = 10^5\times4\times10^{-3} = 400J$

From $A\longrightarrow B, W_2 = \frac{2.2\times10^5\times4\times10^{-3}}{2} = 440J$

$\frac{W_1}{W_2} = \frac{400}{440} = \frac{10}{11}$

31.

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All the elements are in parallel

$$\int \frac{1}{dr} = \int_{R_1}^{R_2} \frac{tdx}{\rho \pi x}$$

$$\frac{1}{r} = \frac{t}{\pi \rho} \ln\left(\frac{R_2}{R_1}\right)$$

$$\text{Resistance} = \frac{\pi}{t \ln\left(\frac{R_2}{R_1}\right)}$$

$$t = \frac{V_0 t \ln}{\pi \rho} \ldots\ldots\ldots\ldots\ldots(1)$$

$\left(-e\vec{E}\right)$ will be inward direction in order upto provide centripetal acceleration.

Therefore electric field will be radially outward

$$V_{outer} < V_{inner} \ldots\ldots\ldots\ldots\ldots(2)$$

$$\frac{mV_d^2}{r} = q\vec{E}$$

$$E = \frac{mV_d^2}{qr} \quad \left(I = neAV_d \Rightarrow V_d \alpha i\right)$$

$$\Delta V = \int \vec{E}.\vec{dr}$$

$$\Delta V \alpha V_d^2$$

$$\Delta V \alpha I^2$$

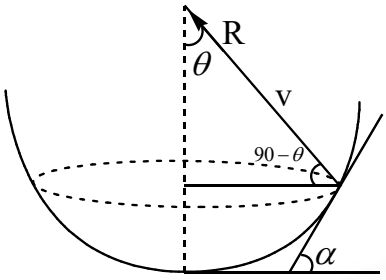
32. $N\cos\theta = mg$

$$N\sin\theta = \frac{mv^2}{x}$$

$$\tan\theta = \frac{v^2}{gx} = \frac{v^2}{gR\sin\theta}$$

$$mvx = \frac{nh}{2\pi}$$

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$$mv \times R\sin\theta = \frac{nh}{2\pi}$$

$$m\sqrt{\frac{gR\sin^2\theta}{\cos\theta}}R\sin\theta = \frac{nh}{2\pi} \qquad m\sqrt{gR^3}\,\frac{\sin^2\theta}{\sqrt{\cos\theta}} = \frac{nh}{2\pi}$$

$$\cos\theta \approx 1 \qquad m\sqrt{gR^3} \times \frac{x^2}{R^2} = \frac{nh}{2\pi}$$

$$x = \sqrt{\frac{nh}{2\pi m}\sqrt{\frac{R}{g}}} \qquad v \propto \sqrt{n}$$

$$v = \frac{\sqrt{nh}}{\sqrt{2\pi}\sqrt{m}} \times \sqrt{\frac{nh}{2\pi m}\sqrt{\frac{R}{g}}} = \sqrt{\frac{nh}{2\pi m}\sqrt{\frac{g}{R}}}$$

$$E = mgh + \frac{1}{2}mv^2$$

$$= mg(R - R\cos\theta) + \frac{1}{2}m \times \frac{nh}{2\pi m}\sqrt{\frac{g}{R}}$$

$$= \frac{1}{2}mgR \times \frac{x^2}{R^2} + \frac{1}{2}\frac{nh}{4\pi}\sqrt{\frac{g}{R}} = \frac{mg}{2R} \times \frac{nh}{2\pi m}\sqrt{\frac{R}{g}} + \frac{nh}{4\pi}\sqrt{\frac{g}{R}} \qquad E = \frac{nh}{2\pi}\sqrt{\frac{g}{R}}$$

33. $\vec{\omega} \times \vec{r} + at\,\hat{j} = 0$

$\omega\hat{k} \times (x\hat{i} + y\hat{j}) + at\ \hat{j} = 0$

$\omega x = -at$ .....(1) $\omega y = 0$ .....(2)

but $y = \frac{1}{2}at^2$ ....(3) from (1) & (3) $y = \frac{\omega^2 x^2}{2a}$

34. CONCEPTUAL

35. a volume V of the gas may be assumed squeezed into the flask by surroundings at

P_0 irreversibly hence $P_0V = \Delta U = n\frac{3R}{2}(T - T_0)$

Where $n = \frac{P_0V}{RT_0}$ thus $T = \frac{5T_0}{3}$

36. force of interaction between two parts

$F = \frac{\sigma^2}{2\epsilon_0}$ Projection area $F = \frac{\sigma^2}{2\epsilon_0}\,\pi(R^2 - r^2)$

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37. except vi, vii remaining all

38. x = 5 (i, ii, iv, vi, vii)
y = 4 (i, ii, iv)

39. For He^+ -

$\Delta E = E_1 = E_2 = 10.2eV$

Finally energy of electron in the H – atom $= -13.6 + 10.2 = -3.4\ eV$

K. E of electron $= +3.4eV$

$$\lambda = \sqrt{\frac{150}{V}} = \sqrt{\frac{150}{3.4}} = 6.64\ A^0$$

$$Z = \frac{\lambda}{0.529A} = 4$$

40. $$K_{sp} = \left[M^{2+}\right]\left[S^{2-}\right] = \frac{\left[M^{2+}\right]K_a(H_2S).[H_2S]}{\left[H^+\right]^2}$$

$$\left[H^+\right]^2 = \frac{\left[M^{2+}\right]K_a(H_2S).[H_2S]}{K_{sp}}$$

$X = 3, Y = 2, Z = 4, P = 1$

41. Ksp of MCl_2 in water is 4×10^{-12}

$$S_1 = \sqrt{\frac{ksp}{4}} = \sqrt[3]{\frac{4 \times 10^{-12}}{4}} = 10^{-4}$$

When $CaCl_2$ is added:

Let the molarity of $CaCl_2 = x$

$$MCl_2 \rightleftharpoons M^{2+} + 2Cl^-$$

$$\underset{S_2}{} \quad \underset{2S_2}{}$$

$$CaCl_2 \rightarrow \underset{x}{Ca^{2+}} + \underset{2x}{2Cl^-}$$

$$\left[Cl^-\right] = (2S_2 + 2x) = 2x$$

$$K_{SP} = \left[M^{+2}\right]\left[Cl^-\right]^2$$

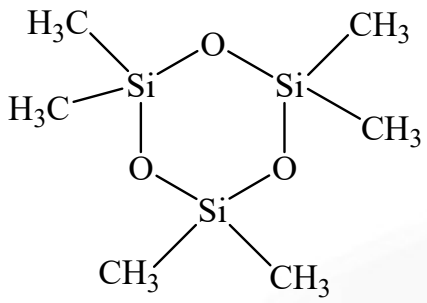
$$K_{SP} = S_2(2x)^2$$

$$S_2 = \frac{K_{SP}}{4x^2} = \frac{10^{-12}}{x^2}$$

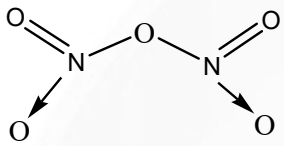
$$\frac{s_1}{s_2} = 4 \times 10^8 \Rightarrow \frac{10^{-4}x^2}{10^{-12}} = 4 \times 10^8 \Rightarrow x = 2$$

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42.

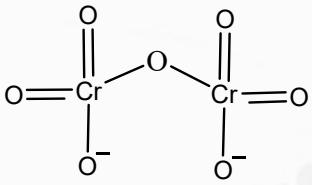


43.



$(a=4)$

(I)



$(x=6)$

(II)

$(x-a)=6-4=2$

44. $x=I_3^-,\ y=IO_3^-$ $\quad IO_3^-:I:+5$ $\quad I_3^-:I:-\frac{1}{3}$

$$\therefore\left[(a\times b)\times(-3)\right]=\left[\left(5\times-\frac{1}{3}\right)(-3)\right]=5$$

45. A = SN^2 product

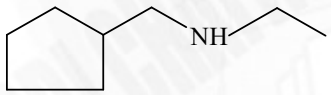
B = SN^i followed by E_2 elimination

C = dehydration followed by aldol condensation

D = ether hydrolysis followed by HI addition

46.

D =



47. Conceptual

48. $\underset{10-3x}{3A} \longrightarrow \underset{\frac{3}{5}\times 4x}{4B} + \underset{\frac{2}{5}\times 7x}{7C}$

Mole of B $=\frac{12x}{5}$

Mole of C $=\frac{14x}{5}$ $\quad \therefore 3x=5\Rightarrow x=5/3$

Mole of B = 4 $\quad$ Mole of C = 14/3

49. A) $en > NH_3$: ligand field strength

B) $ox^{-2} < CN^-$: ligand field strength

$[Fe(ox)_3]^{3-}:\Delta_0=14{,}140cm^{-1}$ $(Huheey)$

$[Fe(CN)_6]^{3-}:\Delta_0=35000cm^{-1}$ $(Huheey)$

C) $[NiCl_2(PPh_3)_2]$: Para: tetrahedral: $Ni^{+2}:d^8$

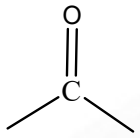
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$CFSE = e^4t_2^4 = [(-0.6\Delta_t)\times 4] + [(+0.4\ \Delta_t)\times 4]$

$= -2.4\Delta_t + 1.6\ \Delta_t \quad = -0.8\ \Delta_t$

D) $Co^{+3} > Co^{2+}$: oxidation state

50. (i) Terminal: $C \equiv O$ (6 terminal CO)



Bridged: (3 bridged CO)

(ii) $[Mn(CO)_6]^+ > [Cr(CO)_6] > [V(CO)_6]^-$: CO bond strength

(iii) $[RhCl(PPh_3)_3]$: square planar

(iv) cis-platin: $[Pt(NH_3)_2Cl_2]$: square planar

51. D, L-Allose

D, L-Galactose

52. D gives enantiomers

53. $X_{He} = \frac{2}{4} = 0.5$

$P_{He} = 6\times 0.5 \quad = 3$ atm

$X_{CH_4} = \frac{1}{4} = 0.25$

$P_{CH_4} = 0.25\times 6 \quad = 1.5$ atm

$X_{SO_2} = \frac{1}{4} = 0.25$

$P_{SO_2} = 0.25\times 6 \quad = 1.5$ atm

rate $\propto \frac{P}{\sqrt{M}}$ $r_{He} \propto \frac{3}{\sqrt{4}} = \frac{3}{2} = 1.5$

$rCH_4 \propto \frac{1.5}{\sqrt{16}} = \frac{1.5}{4} = 0.375$

$r_{SO_2} \propto \frac{1.5}{\sqrt{64}} = \frac{1.5}{8} = 0.1875$

Fraction of He molecule in effused gas stream is

$= \frac{1.5}{1.5 + 0.375 + 0.1875} = 0.7272 = 0.73$

54. i) lead acetate $\xrightarrow{H_2S}$ Black (lead sulphide)

ii) $PbS, CuS \rightarrow Black$

iii) $CrO_2Cl_2 + H_2O \rightarrow H_2CrO_4 + HCl$

iv) $NO_3^- \xrightarrow[+NaOH]{\text{Devardas alloy}} NH_3$

(Devardas alloy: $Zn + Al + Cu$)

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